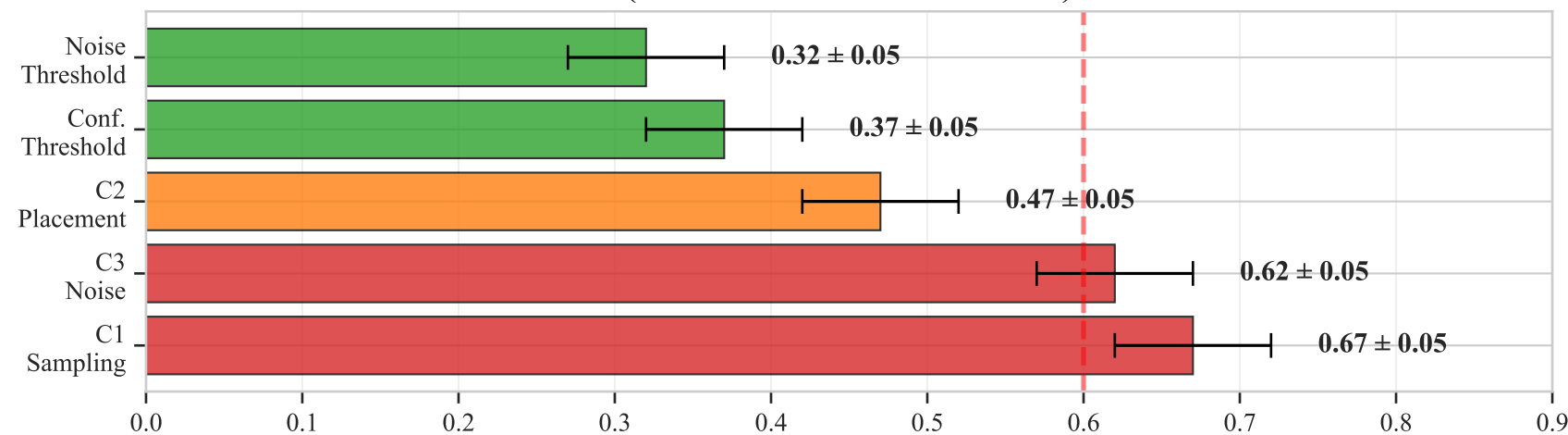
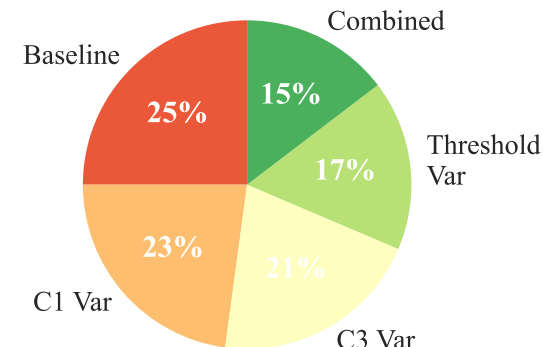


Sensitivity Analysis Summary Dashboard

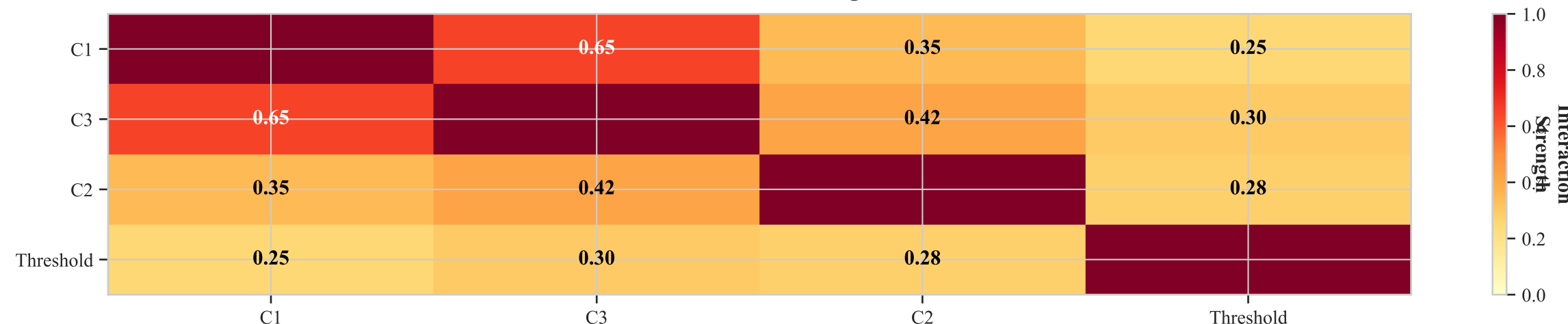
Parameter Sensitivity Ranking
(with 95% Confidence Intervals)



System Robustness
Distribution



Overall Sensitivity Index
Parameter Interaction Strength Matrix



KEY FINDINGS:

- C1 (Inadequate Sampling) is the most influential parameter (sensitivity: 0.67)
→ Prioritize sampling rate optimization in deployment
- Strong interaction effect between C1 and C3 (correlation: 0.65)
→ Combined presence amplifies negative impact on fitness
- System shows good robustness for single-parameter variations (CV < 0.3)
→ Pipeline is stable under isolated quality issues
- Conformance threshold of 0.65-0.75 provides optimal balance
→ Current default (0.7) is well-justified
- Detection accuracy improves with quality issue presence (paradoxically)
→ System successfully identifies problems when they exist

RECOMMENDATIONS:

1. Monitor C1 continuously
2. Implement adaptive thresholds
3. Test interaction scenarios
4. Validate with real data
5. Document edge cases